

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

WAVE MOTION

LECTURE-02 **PHYSICS**

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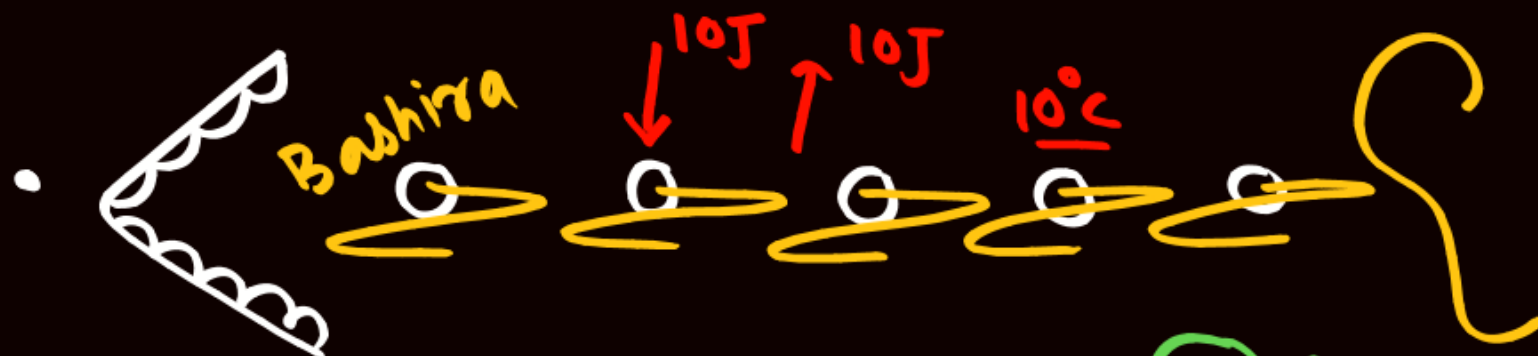
Topics To Be Covered

- 1** Revision dec- 1
- 2** ✓ Standing Waves (5mark)
- 3** ..
- 4** ..
- 5** ..

lecture no. 2

Velocity of sound wave

- Newton
- the propagation of sound wave is a isothermal process
Temp. = constant



$$v^2 \propto \rho \text{ --- (I)}$$

$$\uparrow f = \frac{m}{v} \uparrow$$

$$\downarrow v^2 \propto \frac{1}{\rho} \text{ --- (II)}$$

$$v^2 \propto \frac{P}{\rho}$$

$$v^2 = k \frac{P}{\rho}$$

$$v^2 = 1 \frac{P}{\rho}$$

$$v^2 = \frac{P}{\rho}$$

$$v = \sqrt{\frac{P}{\rho}}$$

$$v = \sqrt{\frac{10^5}{1.23}} = 280 \text{ m/s} \quad \times$$

$$P = 10^5 \text{ Pa}$$

$$\rho = 1.23$$

$$\boxed{k=1}$$

Constant.

①

Laplace Correction of Speed of Sound.

- Laplace
- Propagation of sound is very fast ↑↑
no heat exchange

* Adiabatic Process



Nazia

$$v^2 \propto p \text{ — ①}$$

$$v^2 \propto \frac{1}{\rho} \text{ — ②}$$

$$v^2 \propto \frac{p}{\rho}$$

$$v^2 = \gamma \frac{p}{\rho}$$

$$v = \sqrt{\gamma \frac{p}{\rho}}$$

$$v = \sqrt{\frac{7}{5} \times \frac{10^5}{1.23}}$$

$$v = 331 \text{ m/s} \checkmark$$

$$337 \text{ m/s}$$

$$343 \text{ m/s}$$

Heat Capacity Ratio ② ✓

$\gamma \rightarrow$ Adiabatic constant

$$\gamma = \frac{C_p}{C_v} = \frac{7}{5}$$

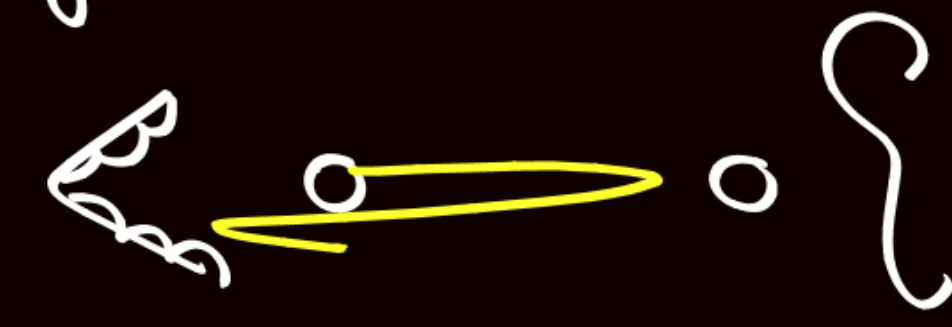
Diatom gas

$$\gamma = 7/5$$

Factor on which speed of Sound depend

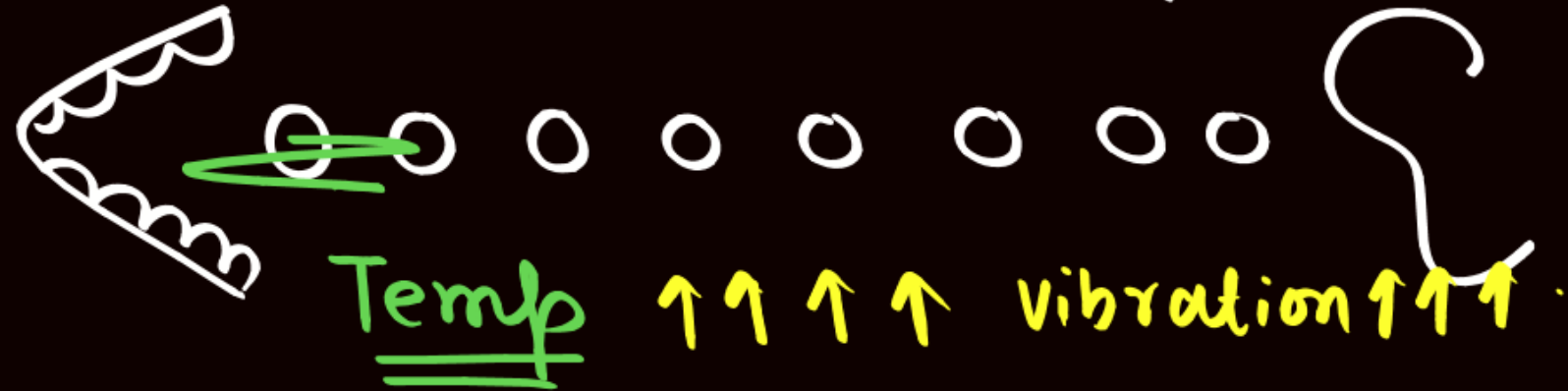
③

① ^{→ state} nature of medium

Solid	liquid	gas
 Bashir	 Bashir	
$V_{solid} >>> V_l >>> V_g$		
Steel 5960 m/s	Water $V = 1480 \text{ m/s}$	Air $V = 331 \text{ m/s}$ $= 341 \text{ m/s}$

② Temperature

on increasing the temperature



Speed of sound ↑↑↑↑↑↑

↑ limited

Temp Speed of sound

Air (15°C) 331 m/s

Air (20°C) 341 m/s

Air (25°C) 345 m/s

④ ✓

(3) Humidity

(presence of water in Air)

⑤



$$f \uparrow \uparrow (\text{air}) \quad V \downarrow \downarrow$$

$$V = \sqrt{\gamma \frac{P}{\rho}}$$

$$\rho_{\text{water}} \downarrow \downarrow \downarrow$$

$$V' = \sqrt{\gamma \frac{P}{\rho'}}$$

on increasing humidity in air
the speed of sound $\uparrow \uparrow \uparrow$



④ Pressure

on increasing the pressure $\uparrow\uparrow$ Speed of sound increase

But density = constant

$$\uparrow\uparrow V = \sqrt{\frac{Y \cdot 2P \uparrow\uparrow}{\rho = \text{constant}}}$$

⑤ Density ρ

on increasing the density $\uparrow\uparrow$ Speed of sound decrease

But pressure = constant

$$\downarrow V = \sqrt{\frac{Y \cdot P \rightarrow \text{constant}}{2\rho \uparrow}}$$

Note

$$V = \sqrt{\frac{Y \cdot P \times 2 \uparrow}{\rho \times 2 \uparrow}}$$

$$V = \sqrt{\frac{Y \cdot P \times 2 \uparrow}{\rho \times 2 \uparrow}}$$

if pressure and density increases
at the same time

no effect on speed of sound

⑥

→ neet-2023

$$\tau_p = \frac{m}{v \downarrow}$$

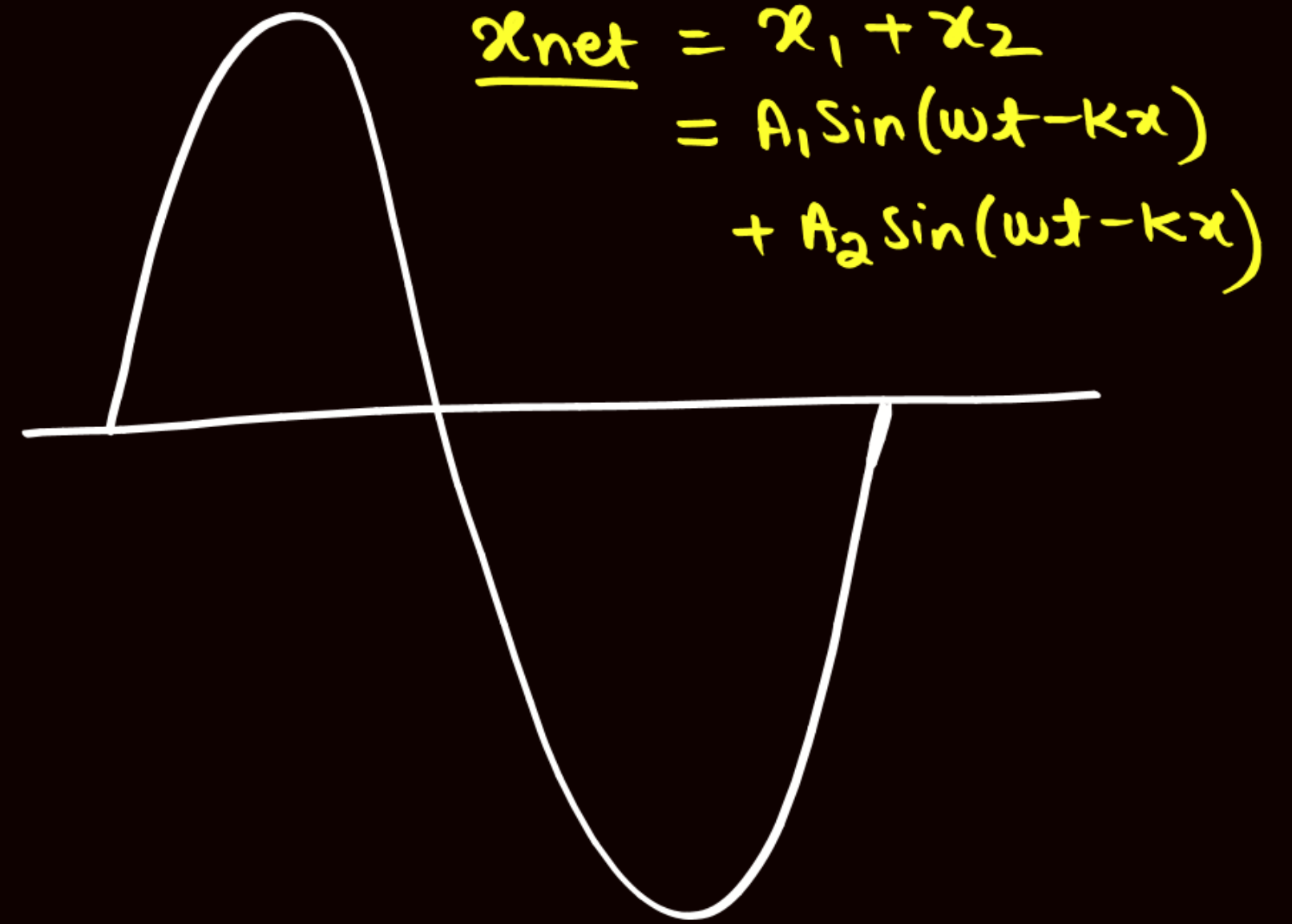
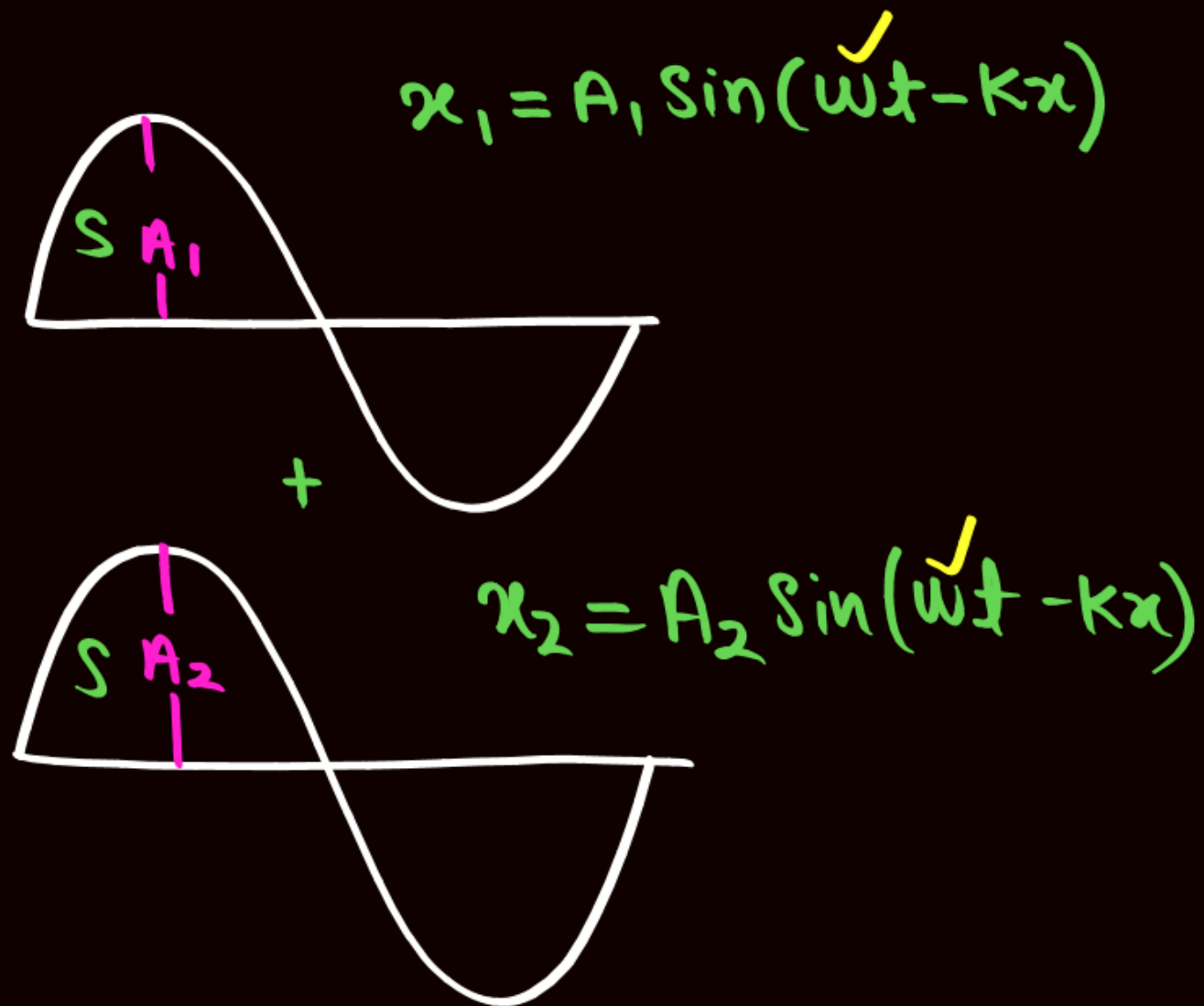
$$\tau_p \propto \frac{1}{v \downarrow}$$

⑦

Superposition of sound wave

(8)

- When one sound wave superimposed on other sound wave with different Amplitude But same frequency



Beats

⑨

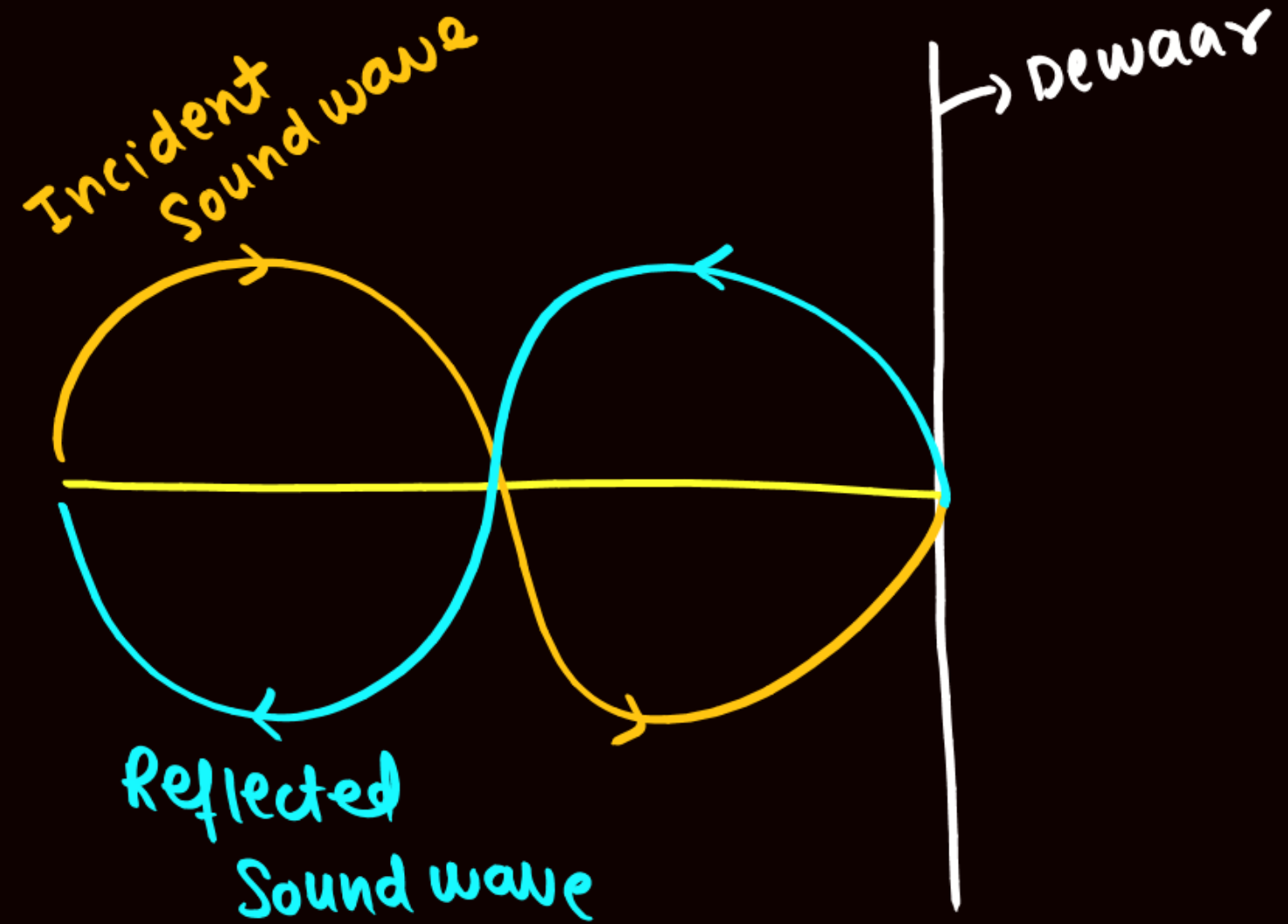
- Beats are formed when two waves of different frequency superimposed

$$x_1 = A_1 \sin(\underline{\omega_1} t - kx) \quad \checkmark$$

$$x_2 = A_2 \sin(\underline{\omega_2} t - kx) \quad \checkmark$$

Reflection of Sound wave

- The Bouncing Back of Sound wave into Same medium



Relationship b/w speed of wave (v), wavelength (λ) and frequency (ν)

$$\text{Speed} = \frac{\text{distance}}{\text{Time}}$$

$$v = \frac{\lambda}{T}$$

$$v = \lambda \times \left(\frac{1}{T} \right) \quad \nu = \frac{1}{T}$$

$$v = \lambda \nu$$

$$v = \lambda \nu$$

$$\begin{array}{c} \uparrow \uparrow \\ \frac{v}{\lambda} = \nu \end{array} \begin{array}{c} \downarrow \downarrow \end{array}$$

$$\begin{array}{c} \downarrow \downarrow \\ \frac{v}{\lambda} = \nu \end{array} \begin{array}{c} \uparrow \uparrow \uparrow \end{array}$$

SUMMARY

$$v = \sqrt{\frac{p}{\rho}}$$

$$v = \sqrt{\gamma \frac{p}{\rho}}$$

v (factor)

SP

Bent

Reflect

HOMework

1. why sound wave / longitudinal wave
are pressure wave

$$v = \sqrt{\frac{\gamma p}{\rho}}$$

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

$$\begin{array}{r} \text{N}_2 + \text{O}_2 \\ + \text{CO}_2 \\ \hline 44 \text{ gm} \\ \hline \text{m} \uparrow \\ \text{g} \uparrow \end{array}$$

$$\begin{array}{r} \text{H}_2\text{O} \\ \hline 18 \text{ gm} \end{array}$$

PW



Pala Wallah